Building and Comparing Two Number Patterns

Building a pattern from its rule, comparing the matching terms of two patterns, and turning two patterns into ordered pairs on a grid.

- A **rule** tells you where to start and what to add each time. The first number you write is term 1.
- **Corresponding terms** are the terms in the same place: term 3 of one pattern goes with term 3 of the other.
- An **ordered pair** puts corresponding terms together as (first pattern's term, second pattern's term), which you can plot on a grid.

1. **Pattern A** follows this rule: start at 0 and add 2 each time. Write the first six terms of Pattern A, then say what its sixth term is.

Answer: __________________

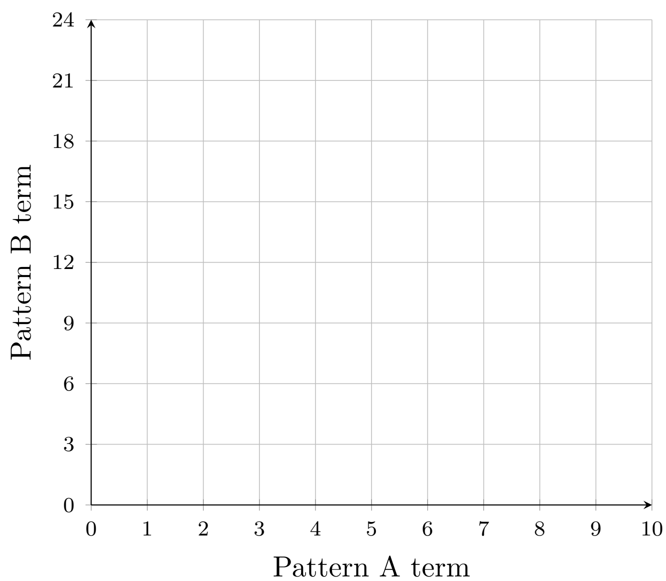
2. **Pattern B** follows this rule: start at 0 and add 6 each time. Write the first six terms of Pattern B, then say what its sixth term is.

Answer: __________________

3. Line up Patterns A and B from problems 1 and 2 so that corresponding terms sit under each other. **(a)** Pattern A's fourth term is 6 and Pattern B's fourth term is 18. Divide to find how many times Pattern B's term is Pattern A's term. **(b)** Pattern A's sixth term is 10. Use the relationship from part (a) to say what Pattern B's sixth term is, without adding all the way along.

Answer: ________________

4. Use Pattern A as the first number and Pattern B as the second number in each ordered pair. **(a)** Write the ordered pairs for terms 1 through 4 and plot them on the grid. **(b)** If both patterns keep going, one ordered pair has 8 as its first number. What is its second number?



Answer: ________________

5. Ana puts 5 dollars in a jar the first week, and adds 3 dollars every week after that. Write Ana's totals for the first four weeks, then find her total in week 8.

Answer: ____________

6. Ben starts the same week with 10 dollars in his jar and adds 6 dollars every week after that. **(a)** Find Ben's total in week 8. **(b)** Ana's week-8 total is 26 dollars. Divide to find how many times Ben's total is Ana's total in that week.

Answer: ____________

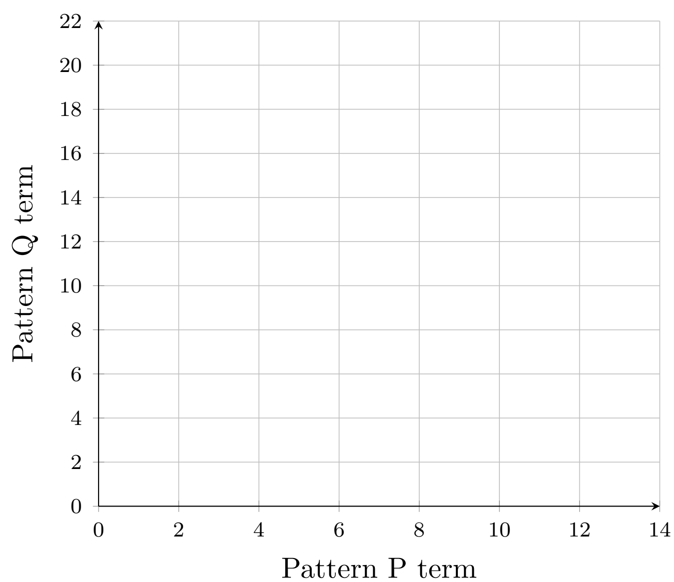
7. Ana's totals for the first five weeks are 5, 8, 11, 14 and 17 dollars. Ben's totals for the same five weeks are 10, 16, 22, 28 and 34 dollars. **(a)** Find the mean of Ana's five totals. **(b)** Find the mean of Ben's five totals. **(c)** Compare your two means: is the second one twice the first?

Answer: ____________

8. **Pattern C:** start at 3 and add 2 each time. **Pattern D:** start at 3 and add 4 each time. Mia says that because Pattern D adds twice as much each time, every term of Pattern D must be twice the corresponding term of Pattern C — so she writes 18 for Pattern D's fourth term. Write out both patterns as far as the fourth term, give the real fourth term of Pattern D, and explain what Mia's rule got wrong.

Answer: ________________

9. **Pattern P:** start at 0 and add 3 each time. **Pattern Q:** start at 0 and add 5 each time. **(a)** Write the ordered pairs (Pattern P term, Pattern Q term) for terms 1 through 5 and plot them on the grid. **(b)** One ordered pair has 6 as its first number. Use your pattern or your graph to give its second number.



Answer: ________________

10. Now write your own pair of rules. Your second pattern's terms must each be four times the corresponding term of your first pattern. Write both rules, list the first five terms of each pattern, and explain how you chose the starting number and the amount to add so that the "four times" relationship works at every single term.

Answer: ________________